

Chapter-2

$$\underline{21} \quad \lim_{n \rightarrow \infty}$$

$$\frac{\left(\frac{1}{1} + \dots + \frac{1}{n} \right)}{1+2+3+\dots+n}$$

(m)
 $\infty \leftarrow \infty$

$$\frac{n^2+1}{(1+n) \cdot n}$$

(m)
 $\infty \leftarrow \infty$

$$= \lim_{n \rightarrow \infty}$$

$$\frac{\frac{n}{2}(n+1)}{n^2+1}$$

$$\frac{\left(\frac{1}{n} + 1 \right) \cdot n}{n^2+1}$$

(m)
 $\infty \leftarrow \infty$

$$= \lim_{n \rightarrow \infty}$$

$$\frac{n^2 \left(1 + \frac{1}{n} \right)}{2n^2 \left(1 + \frac{1}{n^2} \right)}$$

$\frac{1}{2}$

$$= \frac{1+0}{2+(1+0)}$$

$$\frac{1 + \dots + 1}{n^2+1}$$

(m)
 $\infty \leftarrow \infty$

$$= \frac{1}{2} \text{ Ans.}$$

$$\frac{1 + \dots + 1}{n^2+1}$$

(m)
 $\infty \leftarrow \infty$

$$\underline{21} \quad \lim_{n \rightarrow \infty}$$

$$\frac{1^2+2^2+3^2+\dots+n^2}{n^3}$$

$$\frac{1+2+3+\dots+n}{n^3}$$

(m)
 $\infty \leftarrow \infty$

$$= \lim_{n \rightarrow \infty}$$

$$\frac{n(n+1)(n+1)}{6}$$

$$\left(\frac{1}{n^3} \left(\frac{1}{n} + 1 \right) \right) \cdot n$$

(m)
 $\infty \leftarrow \infty$

$$= \lim_{n \rightarrow \infty}$$

$$\frac{n^3 \left(1 + \frac{1}{n} \right) \left(2 + \frac{1}{n} \right)}{6n^3}$$

$\frac{1}{6}$

$$= \lim_{n \rightarrow \infty}$$

$$\frac{\left(1 + \frac{1}{n} \right) \left(2 + \frac{1}{n} \right)}{6}$$

$$= \frac{2}{6}$$

$$= \frac{1}{3}$$

$$7) \lim_{n \rightarrow \infty} \left(\frac{1}{n^2} + \frac{2}{n^2} + \frac{3}{n^2} + \dots + \frac{n}{n^2} \right)$$

$$= \lim_{n \rightarrow \infty} \frac{n(n+1)}{2n^2}$$

$$= \lim_{n \rightarrow \infty} \frac{n^2(1 + \frac{1}{n})}{2n^2}$$

$$= \frac{1}{2}$$

$$8) \lim_{n \rightarrow \infty} \frac{1^3 + 2^3 + 3^3 + \dots + n^3}{n^4}$$

$$= \lim_{n \rightarrow \infty} \frac{n^2(n+1)^2}{4n^4}$$

$$= \lim_{n \rightarrow \infty} \frac{n^2 + 2n + 1}{4n^2}$$

$$= \lim_{n \rightarrow \infty} \frac{n^2(1 + \frac{2}{n} + \frac{1}{n^2})}{4n^2}$$

$$= \frac{1}{4}$$

$$\frac{(\frac{1}{n} + 1)(\frac{1}{n} + 1)}{2}$$

$$\textcircled{C} \lim_{x \rightarrow a} \frac{a - \sqrt{a^2 - x^2}}{x^2}$$

$$= \lim_{x \rightarrow a} \frac{a}{x^2} - \frac{\sqrt{a^2 - x^2}}{x^2}$$

$$= \frac{a}{a^2} - \frac{\sqrt{a^2 - a^2}}{a^2}$$

$$= \frac{1}{a}$$

$$\textcircled{4} \lim_{x \rightarrow 1} \frac{x - \sqrt{2 - x^2}}{2x - \sqrt{2 + x^2}}$$

$$\textcircled{4} \lim_{x \rightarrow 2} \frac{x^2 - 4}{\sqrt{x+2} - \sqrt{3x-2}}$$

$$= \lim_{x \rightarrow 2} \frac{x^2 - 4}{\sqrt{x+2} - \sqrt{3x-2}} \times \frac{\sqrt{x+2} + \sqrt{3x-2}}{\sqrt{x+2} + \sqrt{3x-2}}$$

$$= \lim_{x \rightarrow 2} \frac{(x^2 - 4)(\sqrt{x+2} + \sqrt{3x-2})}{(\sqrt{x+2} - \sqrt{3x-2})(\sqrt{x+2} + \sqrt{3x-2})}$$

$$= \lim_{x \rightarrow 2} \frac{(x+2)(x-2)(\sqrt{x+2} + \sqrt{3x-2})}{-2(x-2)}$$

$$= \lim_{x \rightarrow 2} \frac{(x+2)(\sqrt{x+2} + \sqrt{3x-2})}{-2}$$

$$= \frac{4(2+2)}{-2} = -8$$

$$\lim_{x \rightarrow 1} \frac{x - \sqrt{2-x^2}}{2x - \sqrt{2+2x^2}} = \frac{0}{0} \quad \text{L'Hôpital's Rule}$$

$$= \lim_{x \rightarrow 1} \frac{(x - \sqrt{2-x^2}) (x + \sqrt{2-x^2}) (2x + \sqrt{2+2x^2})}{(2x - \sqrt{2+2x^2}) (2x + \sqrt{2+2x^2}) (x + \sqrt{2-x^2})}$$

$$= \lim_{x \rightarrow 1} \frac{(x^2 - 2 + x^2) (2x + \sqrt{2+2x^2})}{(4x^2 - 2 - 2x^2) (x + \sqrt{2-x^2})}$$

$$= \lim_{x \rightarrow 1} \frac{(2x^2 - 2) (2x + \sqrt{2+2x^2})}{(2x^2 - 2) (x + \sqrt{2-x^2})}$$

$$= \lim_{x \rightarrow 1} \frac{2x + \sqrt{2+2x^2}}{x + \sqrt{2-x^2}} = \frac{0+2}{0+1} = 2$$

$$= \frac{2+2}{1+1} = 2$$

2 2

$$\begin{aligned}
 \text{21) } \lim_{x \rightarrow \infty} \{ \sqrt{x-1} - \sqrt{x} \} &= \lim_{x \rightarrow \infty} \frac{(\sqrt{x-1} + \sqrt{x})(\sqrt{x-1} - \sqrt{x})}{\sqrt{x-1} + \sqrt{x}} \\
 &= \lim_{x \rightarrow \infty} \frac{x-1-x}{\sqrt{x-1} + \sqrt{x}} \\
 &= 0
 \end{aligned}$$

$$\begin{aligned}
 \text{22) } \lim_{x \rightarrow a} \frac{x^{9/2} - a^{9/2}}{\sqrt{x} - \sqrt{a}} &= \lim_{x \rightarrow a} \frac{(\sqrt{x})^9 - (\sqrt{a})^9}{\sqrt{x} - \sqrt{a}} \\
 &= \frac{9(\sqrt{a})^{9-1}}{1} \\
 &= 9a^4
 \end{aligned}$$

$$\begin{aligned}
 \text{20) } \lim_{x \rightarrow \infty} \{ \sqrt{x+1} - \sqrt{x} \} &= \lim_{x \rightarrow \infty} \frac{(\sqrt{x+1} - \sqrt{x})(\sqrt{x+1} + \sqrt{x})}{\sqrt{x+1} + \sqrt{x}} \\
 &= \lim_{x \rightarrow \infty} \frac{x+1-x}{\sqrt{x+1} + \sqrt{x}} \\
 &= 0
 \end{aligned}$$

$$\begin{aligned}
 \text{26) } \lim_{n \rightarrow \infty} \frac{(3^n + 2^n)^{1/n}}{1 + (2/3)^n} &= \lim_{n \rightarrow \infty} \frac{3 + 2(2/3)^n}{1 + (2/3)^n} \\
 &= \frac{3 + 0}{1 + 0} = 3
 \end{aligned}$$

$$\begin{aligned}
 \text{23) } \lim_{x \rightarrow a} \frac{x^{5/2} - a^{5/2}}{\sqrt{x} - \sqrt{a}} &= \lim_{x \rightarrow a} \frac{(\sqrt{x})^5 - (\sqrt{a})^5}{\sqrt{x} - \sqrt{a}} \\
 &= 5(\sqrt{a})^{5-1} \\
 &= 5a^2
 \end{aligned}$$

$$\underline{28)} \lim_{x \rightarrow 0} \frac{e^x - 1}{x}$$

$$= \lim_{x \rightarrow 0} \frac{(1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots) - 1}{x}$$

$$= \lim_{x \rightarrow 0} \frac{x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots}{x}$$

$$= \lim_{x \rightarrow 0} \left(1 + \frac{x}{2!} + \frac{x^2}{3!} + \dots \right)$$

$$= 1$$

$$\underline{29)} \lim_{x \rightarrow 0} \frac{a^x - 1}{x}$$

$$= \lim_{x \rightarrow 0} \frac{(1 + x \ln a + \frac{(x \ln a)^2}{2!} + \dots) - 1}{x}$$

$$= \lim_{x \rightarrow 0} \left(\ln a + \frac{x (\ln a)^2}{2!} + \dots \right)$$

$$= \ln a$$

$$\underline{26)} \lim_{x \rightarrow 0} \frac{(1+x)^n - 1}{x}$$

$$= \lim_{x \rightarrow 0} \frac{1 + nx + \frac{n(n-1)}{2}x^2 + \dots - 1}{x}$$

$$= \lim_{x \rightarrow 0} \left(n + \frac{n(n-1)}{2}x + \dots \right)$$

$$\underline{29)} \lim_{x \rightarrow 0} \frac{\sin^{-1} x}{x}$$

$$= 1$$

$$\underline{26)} \lim_{x \rightarrow 0} \frac{\tan^{-1} x}{x}$$

$$= 1$$

$$\frac{e}{e} = 1$$

lim
Hence

$$1 \cdot 0 = 0$$

26) $\lim_{x \rightarrow 0}$

$$\frac{\tan x - \sin x}{x^3}$$

$$= \lim_{x \rightarrow 0} \frac{\tan x (1 - \cos x)}{x^3}$$

$$= \lim_{x \rightarrow 0} \frac{\tan x}{x} \times \lim_{x \rightarrow 0} \frac{12 \sin^2 x/2}{2}$$

$$= 1 \times \lim_{x \rightarrow 0} \left(\frac{\sin x/2}{x/2} \right)^2 \times \frac{1}{2}$$

$$= 1 \times 1^2 \times \frac{1}{2} \left(\pi + \frac{\pi}{5} \right)$$

$$> \frac{1}{2}$$

$$\text{Q21] } f(x) = \begin{cases} \tan x & \text{when } x < \frac{\pi}{2} \\ 3 - \frac{\pi}{2} & \text{when } x = \frac{\pi}{2} \\ \frac{x^3 - \frac{\pi^3}{8}}{x - \frac{\pi}{2}} & \text{when } x > \frac{\pi}{2} \end{cases}$$

when $x > \frac{\pi}{2}$, $f(x) = \frac{x^3 - \frac{\pi^3}{8}}{x - \frac{\pi}{2}}$

$$\therefore \lim_{x \rightarrow \frac{\pi}{2}^+} f(x) = \lim_{h \rightarrow 0} f\left(\frac{\pi}{2} + h\right)$$

$$= \lim_{h \rightarrow 0} \frac{\left(\frac{\pi}{2} + h\right)^3 - \frac{\pi^3}{8}}{\frac{\pi}{2} + h - \frac{\pi}{2}}$$

(L'Hôpital's rule) $\lim_{h \rightarrow 0} \frac{\frac{3\pi^2}{4}h + \frac{3\pi}{2}h^2 + h^3}{h}$

$$= \lim_{h \rightarrow 0} \frac{3\pi^2}{4} + \frac{3\pi}{2}h + h^2$$

$$= \frac{3\pi^2}{4}$$

when $x < \frac{\pi}{2}$, $f(x) = \tan \frac{x}{2}$

$$\therefore \lim_{x \rightarrow \frac{\pi}{2}^-} f(x) = \lim_{h \rightarrow 0} f\left(\left(\frac{\pi}{2} - h\right)\right) > 0$$

$$= \lim_{h \rightarrow 0} \tan\left(\frac{\pi}{4} - \frac{h}{2}\right)$$

$$= \tan \frac{\pi}{4} = 1$$

$$S=1, 1+x = \frac{1-S}{1-S} = 1$$

$\therefore \lim$

$$\lim_{x \rightarrow \frac{\pi}{2}^+} f(x) \neq \lim_{x \rightarrow \frac{\pi}{2}^-} f(x)$$

$\therefore \lim_{x \rightarrow \frac{\pi}{2}} f(x)$ doesn't exist.

For $x > \frac{\pi}{2}$, $f(x) = \tan \frac{x}{2}$

$\delta \in \mathbb{Q}$ (most important)

$$0 < |x - a| < \delta \Rightarrow |f(x) - l| < \epsilon$$

$$\text{Qn) } \lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} = 2$$

$$\text{Qn) } \lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} = 2$$

here, $a = 1$, $f(x) = \frac{x^2 - 1}{x - 1} = x + 1$, $l = 2$

So,

$$0 < |x - 1| < \delta \Rightarrow |f(x) - 2| < \epsilon$$

$$\Rightarrow |x + 1 - 2| < \epsilon$$

$$\Rightarrow |x - 1| < \epsilon$$

So the function holds all the properties of $\delta \in \mathbb{Q}$.

$$\therefore \lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} = 2 \quad [\text{proved}]$$

$$80) \lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2} = 4$$

here, $a = 2$, $f(x) = \frac{x^2 - 4}{x - 2}$ and $l = 4$

$$= x + 2$$

So,

$$0 < |x - a| < \delta \Rightarrow |f(x) - l| < \epsilon$$

$$\Rightarrow 0 < |x - 2| < \delta \Rightarrow |x + 2 - 4| < \epsilon$$

$$\Rightarrow |x - 2| < \epsilon$$

$$\therefore \delta = \epsilon$$

So the function holds all the properties of $\delta \in$.

$$\therefore \lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2} = 4 \quad [\text{Proved}]$$

$$\underline{87)} \quad \lim_{x \rightarrow 2} \frac{2x^2 - 8}{x - 2} = 8$$

$$\text{here, } a = 2, \quad f(x) = \frac{2(x^2 - 4)}{x - 2}, \quad l = 8$$

$$= 2x + 4$$

$$\text{So, } 0 < |x - 2| < \delta \Rightarrow |2x + 4 - 8| < \epsilon$$

$$\Rightarrow |2x - 4| < \epsilon$$

$$\Rightarrow |2(x - 2)| < \epsilon$$

$$\Rightarrow |x - 2| < \frac{\epsilon}{2}$$

$$\Rightarrow |x - 2| < \frac{0.1}{2}$$

$$\Rightarrow |x - 2| < 0.05$$

$$\underline{88)} \quad \lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3} = 6$$

881 $\lim_{x \rightarrow 1} \frac{x^3 - 1}{x - 1} = 3$

$a = 1$, $f(x) = \frac{x^3 - 1}{x - 1}$

$\frac{x^3 - 1}{x - 1} = \frac{(x - 1)(x^2 + x + 1)}{x - 1}$

$= x^2 + x + 1$

So, $0 < |x - 1| < \delta \Rightarrow |x^2 + x + 1 - 3| < \epsilon$

$\Rightarrow |x^2 + x - 2| < \epsilon$

$= |(x - 1)(x + 2)| < \epsilon$

$= |x + 2| |x - 1| < \epsilon$

As, $x \rightarrow 1$, so, $0 < x < 2$ and $2 < x + 2 < 4$

~~$0 < |x - 1| < \delta \Rightarrow$~~

~~$0 < |x + 2| |x - 1| < 4\delta \Rightarrow |x + 2| |x - 1| < \epsilon$~~

~~$4\delta = \epsilon$~~

~~$\Rightarrow \delta = \epsilon/4$~~

$\therefore$ the function holds all the properties of $\delta \in \mathbb{R}$ if we put $\delta = \frac{\epsilon}{4}$

$$\therefore \lim_{x \rightarrow 1} \frac{x^3 - 1}{x - 1} = 3$$

$$\text{88) } \lim_{x \rightarrow 1} \frac{x-1}{x^2+2x-3} = \frac{1}{4}$$

$$\begin{aligned} a=1, \quad f(x) &= \frac{x-1}{x^2+2x-3}, \quad L = \frac{1}{4} \\ &= \frac{x-1}{(x-1)(x+3)} \\ &= \frac{1}{x+3} \end{aligned}$$

$$\text{So, } 0 < |x-1| < \delta \Rightarrow \left| \frac{1}{x+3} - \frac{1}{4} \right| < \epsilon$$

$$\Rightarrow \left| \frac{4-x-3}{4(x+3)} \right| < \epsilon$$

$$\Rightarrow \left| \frac{1-x}{4(x+3)} \right| < \epsilon$$

As, $x \rightarrow 1$ so let $0 < x < 2$ then

$$3 < x+3 < 5$$

$$\Rightarrow \frac{1}{5} < \frac{1}{x+3} < \frac{1}{3}$$

$$\therefore \left| \frac{1-x}{4(x+3)} \right| < \frac{\delta}{12} = \epsilon$$

$$\therefore \delta \leq 12\epsilon$$

So ~~of~~ the function ~~will~~ hold all the properties of $S \in$ if we put $S \leq 12 \epsilon \cdot \frac{1}{p} = \frac{1-x}{\epsilon-x+\epsilon x}$ ~~now~~ $1 \leftarrow x$

$$\therefore \lim_{x \rightarrow 1} \frac{x-1}{x^2+2x-3} = \frac{1}{4} \quad \text{L'Hôpital's Rule}$$

$$\frac{1-x}{(\epsilon+x)(1-x)} =$$

$$\frac{1}{\epsilon+x} =$$

$$\exists \delta > 0 \text{ such that } \left| \frac{1}{p} - \frac{1}{\epsilon+x} \right| < \delta \text{ whenever } |x-1| < \delta$$

$$\exists \delta > 0 \text{ such that } \left| \frac{\epsilon-x-p}{(\epsilon+x)p} \right| < \delta$$

$$\exists \delta > 0 \text{ such that } \left| \frac{x-1}{(\epsilon+x)p} \right| < \delta$$

now $\delta > x > 0$ for $0 < x < 1$

$$\delta > \epsilon+x > \epsilon$$

$$\frac{1}{2} < \frac{1}{\epsilon+x} < \frac{1}{\epsilon}$$

$$\delta = \frac{\epsilon}{15} > \left| \frac{x-1}{(\epsilon+x)p} \right|$$

$$\delta \leq 1 \leq \epsilon$$